

Research Review Paper

Large Language Models Enabled Recommendation Systems: A Systematic Review

Course: INFO 556 Information Retrieval and Web Search

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Abstract

With the rapid advancement in large language models (LLMs) and the rise of artificial intelligence (AI) developments, it has become imperative to utilise the scale and utility of such models to enhance the current recommendation systems. Traditional recommendation systems are heavily based on collaborative filtering, which matches the item-user pairs to recommend the most relevant products or services, depending on the context and business use cases. However, deep generative AI models can now represent the complex relationships between multimedia data, such as text, images, videos, and audio, thereby enhancing the relevance of retrieval items compared to traditional user-item recommendations. This comprehensive review covers the recommendation system (RS) with large language models, focusing on the current work on such generative recommendation systems (GenRecSys). This review surveys the existing methods along with their limitations to understand the scope of such systems. I also include the extensive review strategy and ethical considerations of such a review study to provide a valuable guide to practitioners and researchers.

Introduction

Recommendation systems have consistently been part of our daily lives since the era of e-commerce and streaming platforms. Platforms such as Hulu, Amazon, Netflix, etc, all use the recommendation system to personalise the user experience. In the last few years, large language models have advanced a lot in their capabilities and accuracy in generating user-centric answers. Traditional recommendation systems have been using item or user-based similarity, but they still can be vague and fail to take user interest and personalised context into account. Algorithm advancements have happened over neural networks and machine learning, but still, these advancements can not take multi-dimensional, multi-media data of today's era into consideration, which results in significant accuracy losses (Nawara et al. 2025). As advancement in natural language processing (NLP) happens, LLMs are going to become highly capable of understanding user context based on input queries and other input data in other media as well. These transformer based models can understand the product details, such as book titles, author, publication and public reviews, so that when other users search or find the relevant books, LLM LLM-powered recommendation systems can identify the top relevant books with high precision compared to traditional systems (Zihuai Zhao et al. 2024). There have been various experiments and paradigms since 2021 in the domain of the LLM-enabled RecSys that continue to become mainstream. This review includes various research review papers in this domain, which include the foundational paradigms where LLM is used for zero-shot and few-shot learning to recommend books or products. This review identifies the studies between 2021 and 2025 to analyse the progression of the usage of LLMs in RecSys (Gao et al., 2023).

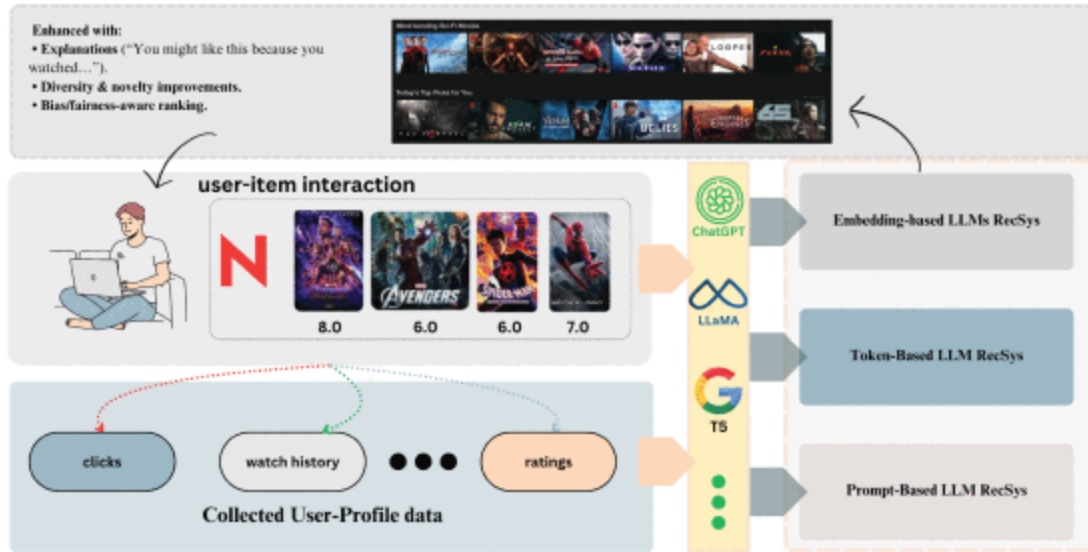


Figure 1: A demonstrative figure of the LLM-based recommendations system (Naware et al. 2025)

The major advantage of incorporating the LLMs into recommendation systems lies in their ability to extract high-quality representations of textual and multimedia features that can be used to extract the extensive knowledge encoded from such data. This encoding strength, coupled with the chat and generating capability of the LLM, can recommend highly accurate and precise outputs to the users (Wu et al. 2024).

Recommendation systems can also be integrated with discriminative models. However, the evolution of the AI learning paradigms and frameworks, generative AI have gained prominence compared to discriminative models. Generative models are highly customizable and context-aware, which fulfils the objective of the personalised recommendations and improves the traditional systems (Wu et al. 2024). This review lays out the detailed review strategy and the models or frameworks used for the large language model-enabled recommendation systems.

Research Questions

RQ1: What is the purpose of utilising the large language models in current recommendation systems review studies?

RQ2: What methods/models or frameworks are being used in LLM-enabled recommendation systems review studies?

RQ3: What are some limitations of the review study on such LLM-enabled recommendation systems?

RQ4: What are some of the ethical considerations one should keep in mind for such review studies?

Literature Review

The advancements of the large language models in information retrieval have necessitated a re-evaluation of the traditional recommendation architectures. While traditional systems are described as narrow experts that rely heavily on collaborative filtering, recent literature highlights the paradigm shift towards the generative models based zero zero-shot or few-shot reasoning, the systems that personalise the conversation and recommend items based on the context of the user session. This review synthesises the recent key survey papers (Lin et al. 2025, Wu et al. 2024, Deldjoo et al. 2024, Zhao et al. 2024, Naware et al. 2025) to categorise the emerging design of the LLM-enabled recommendation systems.

A primary focus of the literature of establishing the taxonomy for how LLMs are integrated into the RecSys pipeline and what the objectives of such methods are in the pipeline. Lin et al. (2025) propose a framework that dissects these systems on two dimensions, which include “Where” the LLM is placed in the pipeline and “How” it is trained.

(Lin et al. 2025) Identifies that LLMs are rarely used in isolation; instead, they are used in functions such as feature encoders to extract the textual side of the information. A scoring or relevancy ranking function can be used as a user interaction layer to manage the dialogue in the systems.

Another dimension, which is based on fine-tuning of the LLM based on prompting techniques, design of the prompts and further expansion of the pre-training, fine-tuning paradigm in the context of recommendation systems (Zhao et al. 2024).

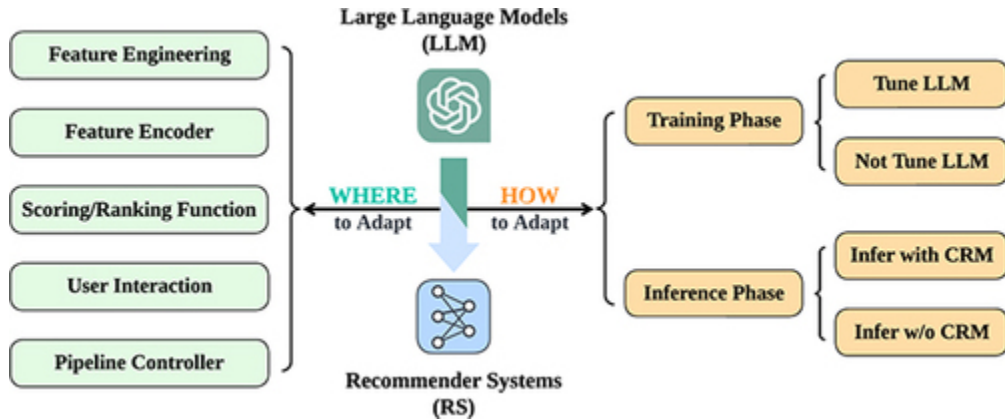


Figure 2: Decomposition of where and how the LLMs are used in RS (Lin et al. 2025)

Wu et al. (2024) provide a brief discussion on Discrimination LLM for the recommendation (DLLM4Rec) and generative LLM recommendation systems (GLLM4Rec).

In the DLLM4Rec design, the LLM acts as an auxiliary component to traditional discriminative models; it enhances the semantic understanding of the queries or items, but the final output remains the retrieval score of the classification model. While GLLM4Rec represents the design change in which LLM is the sole recommendation based on the generative models, it generated the recommendations directly in natural language. As it uses LLMs as a recommender, it suffers from hallucinations.

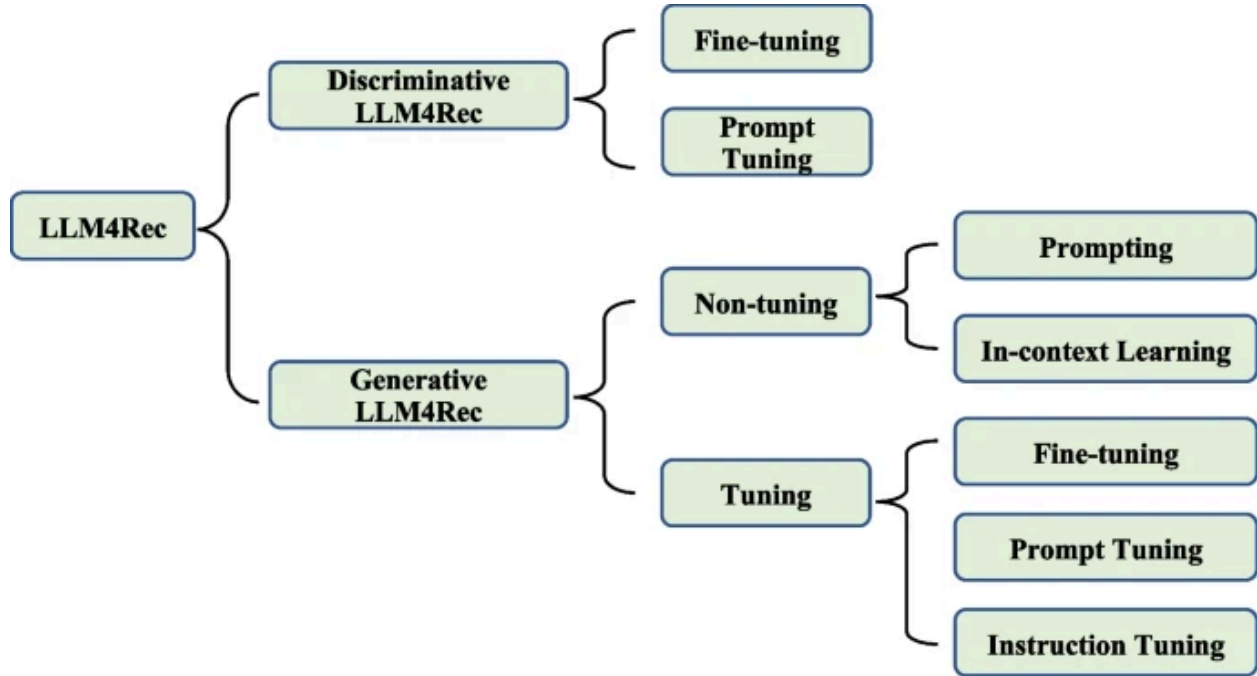


Figure 3: A taxonomy of the LLM models on the recommendations (Wu et al. 2024)

Another paradigm that was used to model the LLM-powered recommendation system is designed based on the embeddings and tokens of the natural language, coupled with the knowledge systems of the databases (Wu et al. 2024).

Generative LLM-based RS heavily depend on the Tuning of the models based on the context and knowledge system being talked about. “Fine-tuning”, “Prompt-Tuning”, “Instruction-tuning”, and “In-context Learning” are some of the techniques for the LLM-enabled RS (Wu et al. 2024).

Deldjoo et al. (2024) broaden the scope beyond just LLM to what they term generative recommender systems. It talks about the defining characteristics of modern designs is the shift from the “interaction-driven” to “multi-model” recommendations. This study emphasises new tasks enabled by these designs, such as Explanation generation and user profiling.

LLM-enabled recommendation systems can mainly operate on the embeddings-based models, token-based models and prompt or prompt engineering-based models to tune and adapt to the recommendation task at hand. LLMs are also used as augmented objects with recommendation strategies in hybrid systems where LLMs, along with other databases, models, multimedia and prompts, are taken into consideration to develop systems (Naware et al. 2025).

Review Methodology

This section outlines the systematic review approach used to identify, evaluate, and select relevant review papers focusing on evaluation metrics for Large Language Model (LLM)-powered recommendation systems.

Search strategy

A comprehensive literature search was conducted to identify the review research papers from 2021 to 2025, reflecting the growing and emerging adoption of LLMs in recommender systems. The search utilised the two primary databases. Google Scholar for broader coverage of the literature, while the University of Arizona Library for high-impact, peer-reviewed articles and journals.

The initial broad search on Google Scholar yielded approximately 18,000 results, indicating a vast volume of general literature on LLMs and recommendations. A focused secondary search was performed on the University of Arizona Library sites to filter for high-quality academic reviews. The selection process followed a "funnel" approach, narrowing down thousands of initial hits to a final set of 5 key review papers, which were reviewed and included in the qualitative synthesis of literature on LLM-enabled recommendation systems.

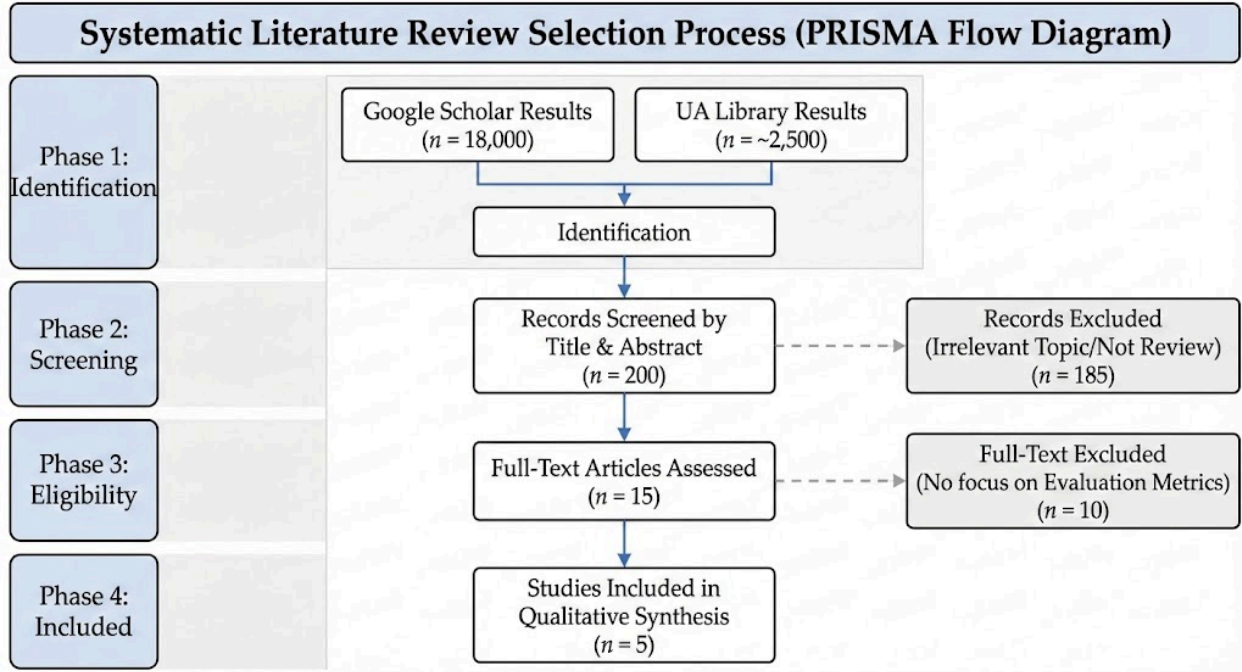


Figure 4: Systematic Literature Review process diagram

Inclusion and Exclusion Criteria

The included topics were explicitly based on the LLM-based recommendation systems. Document types were review papers, systematic surveys, and meta-analyses, with the language of the papers being English. The timeframe of the publication was between 2021 and 2025. Since the Large language model became popular after 2021, I included papers from that year. The excluded topics were traditional neural network algorithm-based RS systems, such as CNN, RNNs, etc. Moreover, Original research papers from Arxiv, reports, editorials or industry articles and non-English papers were excluded. The study also excludes papers published before the year of 2021.

Search terms

A study was done through the keyword search on the large University of Arizona and Google Scholar databases with terms such as “LLM recommendation system REVIEW” and “LLM recommendation systems review”. The timeframe, language and document type filters were applied for the review study as mentioned in the search strategy.

Ethical considerations

This review study can have multiple ethical considerations that must be included in the study. One such consideration is bias propagation by analysing the review papers based on the gender,

race and geography for the LLM-enabled recommendation systems. Another one is privacy and data leakage from the papers' research studies. Citation biases and fairness are other ethical considerations that may impact the review study.

Limitations

The review study proposes several limitations for the review paper, such as the rapid obsolescence of the LLM research, and older papers might be considered outdated as of now. Some papers might be relevant that were published before 2021 are not part of this study. This review study is also keyword sensitive, such as “LLM”, “Recommendation system”, which can exclude the study of works like “Generative AI” or “Transformers-based recommendation system”. The reliance on English-language papers and major indexes (Google Scholar/UA Library) may exclude significant non-English contributions.

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Appendix

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